Data Domain Standard
WyIT.2001-S4

Document Owner	Department of Enterprise Services, Data Privacy
Document Contact	Pat Potter, Deputy Chief Data Officer
Document Approver	Wyoming Data Governance Committee (WDGC)
Effective Date	September 12, 2025
Next Review Date	April 1, 2027
Document Version	2026-09.1
Associated Policy	WyIT 2001, Information Classification

Introduction:

This standard establishes the structural and organizational requirements and characteristics that data domains must meet to support Data Owner management activities as defined in WyIT 2001-S1, Data Owner Standard.

This standard applies to all Executive Branch agencies, boards, and commissions staff (collectively referred to as "agencies"). This standard is also applicable to consultants, affiliates, third party vendors and temporary employees that use or store data.

Authority

This standard is promulgated under the authority of Wyoming State Statute 9-2-2906 and its subsections, the Department of Enterprise Technology Services is required to adopt and enforce information technology standards, and promote efficient use of information technology throughout state government.

Specifically, this policy is further promulgated under the authority of the below listed Wyoming State Statute subsections.

- Implements and maintains an information technology governance program for state agencies (W.S 9-2-2906,(b), (ii));
- Establishes and enforces data security and privacy policies and standards for the state data infrastructure Establishes review criteria for the agencies information technology projects, procurements and purchases. (W.S 9-2-2906,(b), (iii)); and

Adopts information technology policies and standards and ensures agency compliance with those policies and standards (W.S. 9-2-2906,(b), (vi)).

Audience

This policy applies to all Executive Branch agencies, boards, and commissions staff (collectively referred to as "agencies"). This policy is also applicable to consultants, affiliates, third party vendors and temporary employees that use or store state data.

Definitions

Key Definitions, Acronyms, and Terms

Primary Duties:

1. **DUTY DESCRIPTION**
   - 1.1. Data domains shall have clearly defined boundaries that logically group related data types, systems, or business functions. Domains may be organized around agency programs, business services, application systems, or data types, and shall encompass both structured data (such as databases, spreadsheets, and organized electronic files) and unstructured data (such as documents, emails, multimedia files, and social media content). Domain boundaries shall minimize governance gaps and overlapping authority while supporting Data Owner management responsibilities.

   - 1.2. Data domains shall be formally documented within agency operational scope, including domain boundaries, included data types, responsible systems, Data Owner assignment, regulatory considerations, and interdependencies with other domains. Domain documentation shall be maintained in a format accessible to Data Owners, Data Stewards, and audit functions.

   - 1.3. Data domains shall maintain classification levels that reflect the highest sensitivity of component data within the domain. Domain classification shall provide the framework for Data Steward classification activities and regulatory compliance requirements.

   - 1.4. Data domains shall have assigned Data Owners with documented authority as specified in WyIT 2001-S1. Domain governance structure shall support clear accountability and decision-making processes.

   - 1.5. Data domains that share data or have operational dependencies shall maintain documented interfaces and coordination mechanisms. Cross-domain activities shall preserve individual Data Owner authority while ensuring consistent governance application.

   - 1.6. Data domains containing regulated information shall be structured to support specific regulatory requirements including HIPAA, FERPA, CJIS, PCI DSS, and IRS 1075. Regulatory domains shall implement enhanced documentation and control requirements.

The below table references federal audit standard requirements for 'data owners'.

Requirement	NIST Cybersecurity Framework (CSF) 2.0	NIST Risk Management Framework (RMF) – SP 800-37 Rev. 2	NIST Privacy Framework Version 1.0
REQ-003	ID.AM-05, PR.AC-01	RA-2, AC-2	ID.IM-P2, CT.PO-P1
REQ-004	GV.RR-02, GV.OC-02	PM-2, PS-2	GV.PO-P3, GV.RR-P1
REQ-005	GV.SC-01, PR.AC-01	CA-3, PM-1	CT.PO-P2, PR.PO-P1
REQ-006	GV.PO-01, PR.AC-01	AC-2, CA-2	GV.PO-P4, CT.PO-P3
REQ-007	GV.RR-01, GV.SC-01	PM-1, PM-2	GV.MT-P1, GV.PO-P1
REQ-008	GV.OC-02, ID.AM-01	PM-5, CA-5	GV.PO-P2, ID.IM-P1
REQ-009	PR.DS-02, GV.SC-01	MP-3, MP-4	PR.DS-P1, PR.DS-P2
REQ-010	DE.CM-01, RS.CO-02	IR-6, IR-8	PR.IR-P1, GV.MT-P2
REQ-011	GV.RR-01, GV.SC-02	PM-2, CA-2	GV.RR-P1, GV.RR-P2

Requirement	NIST Cybersecurity Framework (CSF) 2.0	NIST Risk Management Framework (RMF) – SP 800-37 Rev. 2	NIST Privacy Framework Version 1.0
REQ-003	ID.AM-05, PR.AC-01	RA-2, AC-2	ID.IM-P2, CT.PO-P1
REQ-004	GV.RR-02, GV.OC-02	PM-2, PS-2	GV.PO-P3, GV.RR-P1
REQ-005	GV.SC-01, PR.AC-01	CA-3, PM-1	CT.PO-P2, PR.PO-P1
REQ-006	GV.PO-01, PR.AC-01	AC-2, CA-2	GV.PO-P4, CT.PO-P3
REQ-007	GV.RR-01, GV.SC-01	PM-1, PM-2	GV.MT-P1, GV.PO-P1
REQ-008	GV.OC-02, ID.AM-01	PM-5, CA-5	GV.PO-P2, ID.IM-P1
REQ-009	PR.DS-02, GV.SC-01	MP-3, MP-4	PR.DS-P1, PR.DS-P2
REQ-010	DE.CM-01, RS.CO-02	IR-6, IR-8	PR.IR-P1, GV.MT-P2
REQ-011	GV.RR-01, GV.SC-02	PM-2, CA-2	GV.RR-P1, GV.RR-P2

Audit Controls and NIST Framework Mappings:

These are the standard controls that are from prior audits.

1. **(REQ-001) Domain Structure and Boundary Definition**

a. Implementation. Data domains shall have clearly defined boundaries that logically group related data types, systems, or business functions to minimize governance gaps and support Data Owner management responsibilities.

b. Validation. Verify domain boundaries are documented, logical, and minimize overlapping authority through boundary definition audits and governance coverage assessments.

2. (REQ-002) Domain Documentation Compliance

a. Implementation. Data domains shall maintain standardized documentation including domain scope, data types, systems, Data Owner assignments, regulatory considerations, and interdependencies.

b. Validation. Audit domain documentation for completeness and standardization, verifying all required elements are current and accessible to authorized personnel.

3. (REQ-003) Domain Classification Standards

a. Implementation. Data domains shall maintain classification levels reflecting the highest sensitivity of component data and provide framework for Data Steward classification activities within the domain.

b. Validation. Review domain classification assignments and verify they properly reflect component data sensitivity and support operational classification activities.

4. (REQ-004) Domain Governance Structure Verification

a. Implementation. Data domains shall have assigned Data Owners with documented authority as specified in WyIT 2001-S1 and clear accountability structures for domain governance decisions.

b. Validation. Verify Data Owner assignments are current, authority is documented, and governance structures support effective domain management and decision-making.

5. (REQ-005) Cross-Domain Interface Management

a. Implementation. Data domains with shared data or operational dependencies shall maintain documented interfaces and coordination mechanisms that preserve individual Data Owner authority.

b. Validation. Audit cross-domain interfaces and coordination mechanisms, verifying they facilitate necessary collaboration while maintaining clear authority boundaries.

6. **(REQ-006) Regulatory Domain Structure Compliance**
    a. Implementation. Data domains containing regulated information shall implement enhanced structure requirements to support specific regulatory compliance including HIPAA, FERPA, CJIS, PCI DSS, and IRS 1075
    b. Validation. Verify regulatory domains meet enhanced structural requirements and support regulatory compliance through specialized audits and compliance assessments.

7. **(REQ-007) Domain Lifecycle Management Standards**
    a. Implementation. Data domains shall follow standardized lifecycle processes for domain creation, modification, merging, splitting, and retirement with appropriate Data Owner approval and documentation
    b. Validation. Review domain lifecycle documentation and verify changes follow established processes with proper approvals and impact assessments.

8. **(REQ-008) Domain Integration Standards**
    a. Implementation. Data domains shall align with enterprise governance initiatives while maintaining support for Data Owner management authority and agency-specific operational requirements.
    b. Validation. Assess domain alignment with enterprise standards and verify integration supports rather than conflicts with Data Owner domain management responsibilities.

9. **(REQ-009) Domain Security and Access Standards**
    a. Implementation. Data domains shall implement appropriate security measures and access controls based on domain classification levels and regulatory requirements.
    b. Validation. Verify domain security implementations align with classification levels and regulatory requirements through security control assessments.

10. **(REQ-010) Domain Incident Response Standards**
    a. Implementation. Data domains shall have documented incident response procedures that integrate with agency and enterprise incident response processes while supporting Data Owner incident management responsibilities.
    b. Validation. Review domain incident response procedures and verify they integrate appropriately with broader incident response frameworks.

11. (REQ-011) Domain Reporting and Monitoring Standards

a. Implementation. Data domains shall support governance reporting requirements and monitoring activities that provide visibility into domain health and compliance status.

b. Validation. Verify domain reporting and monitoring capabilities provide appropriate visibility and support governance oversight requirements.

Applicable State Policies, Federal, State or Industry Rules, Laws, and/or Regulations

1. WyIT 2001, Information Classification
2. WyIT 2001-S1, Data Owner Standard
3. WyIT 2001-S2, Data Steward Standard
4. WyIT 2001-S3, Data Custodian Standard
5. WyIT 2001-G1, Information Classification Guide
6. WyIT 2001-GL2, Information Classification Procedures Guide

Compliance & Exceptions:

Compliance

This document supports an established policy, so there are no compliance requirements.

Exceptions

Since standards are not held to a compliance standard, there are no exceptions to this standard.

Review/Approval:

Date Issued/Review Date

This policy shall be reviewed at least annually and updated / reviewed / approved as defined in the Office of Chief Information Officer (OCIO) Policy lifecycle.

The review and update process shall be coordinated by Governance, Risk and Compliance (GRC), completed by the policy owner and approved through the OCIO policy review workflow.

Approval

This policy as signed by the appropriate officer (listed below) is considered approved and enforced, per Wyoming State Statutes (W.S 9-2-2906,(b), (iii)) and (W.S. 9-2-2906,(b), (vi)).

To obtain a signed copy of this document, contact Enterprise Technology Services Governance, Risk and Compliance (ets-ocio-grc@wyo.gov).

Jeffrey Clines
Officer Name

Chief Information Officer
Officer Title

_____
Officer's Signature

_____
Date

Revision History

The current version of this policy, as identified in the Revision History below, supersedes all previous documents listed below.

Version	Change	Author(s)	Date of Change
2026-09.1	ETS Director signature	GRC	September 2026
2025-09.1	Original Document	Pat Potter, Deputy CDO	September 2025